

DSEL

Document-Semantic Entity Lookup

Retrieval System — Performance Report

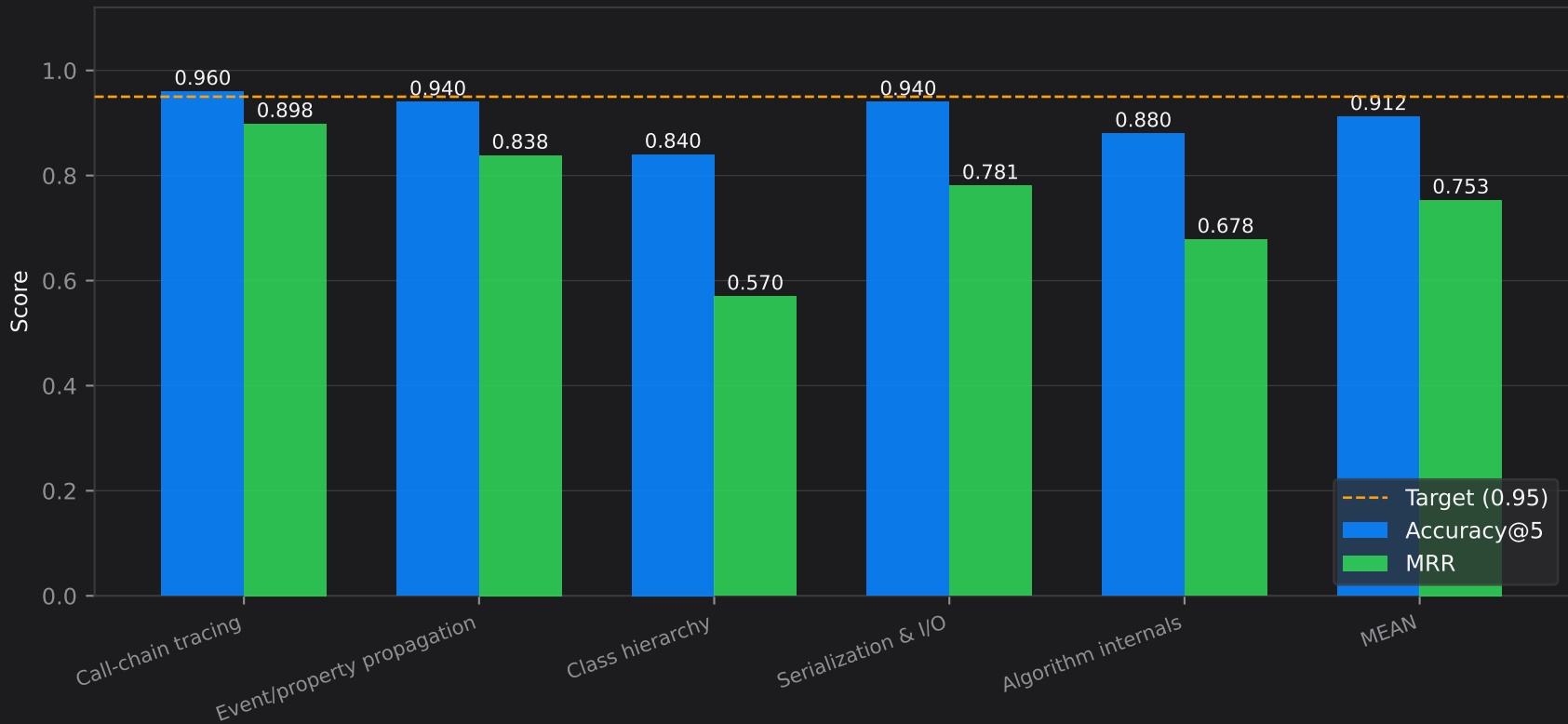
June 08, 2026

FreeCAD codebase · 250-question UMMDB benchmark
SHA-256 hashing embeddings · SQLite vector store · Lexical reranker

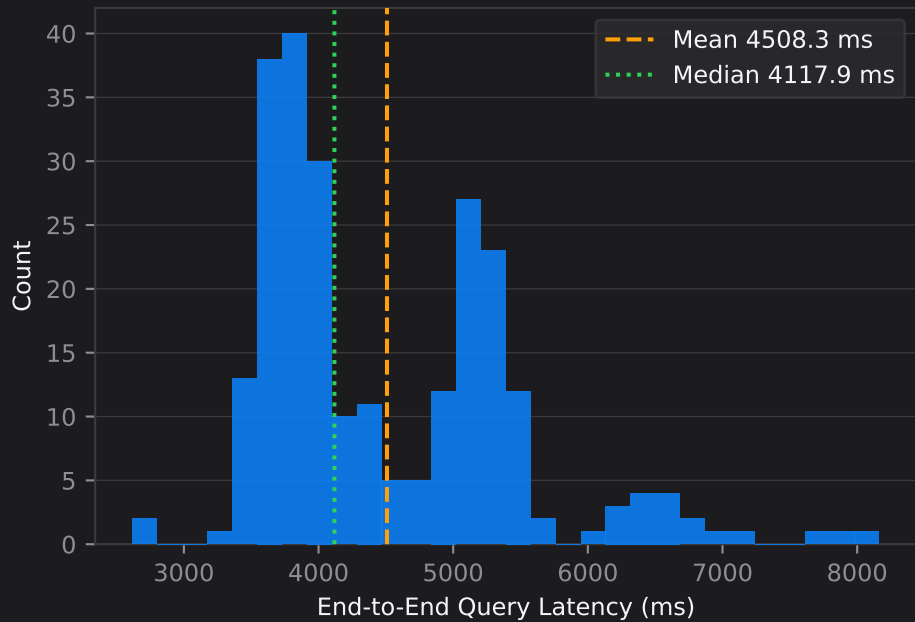
DSEL Retrieval System — Performance Summary

Corpus size	454,005 artifacts		
Benchmark size	250 questions × 5 seeds		
Accuracy@5 (mean)	0.9120		
MRR (mean)	0.7531		
Seed 1: Call-chain tracing	Acc=0.960	MRR=0.898	
Seed 2: Event/property propagation	Acc=0.940	MRR=0.838	
Seed 3: Class hierarchy	Acc=0.840	MRR=0.570	
Seed 4: Serialization & I/O	Acc=0.940	MRR=0.781	
Seed 5: Algorithm internals	Acc=0.880	MRR=0.678	
Mean query latency	4508.34 ms		
Median query latency	4117.92 ms		
P95 query latency	6455.39 ms		
P99 query latency	7459.82 ms		
Hybrid search (mean)	4477.81 ms		
Lexical rerank (mean)	30.41 ms		
Cache build (one-shot)	4617.0 ms		
Amortized encode cost	10.17 μs/artifact		
Embedding backend	SHA-256 hashing (no GPU)		
Reranker	LexicalReranker (term overlap + symbol matching)		

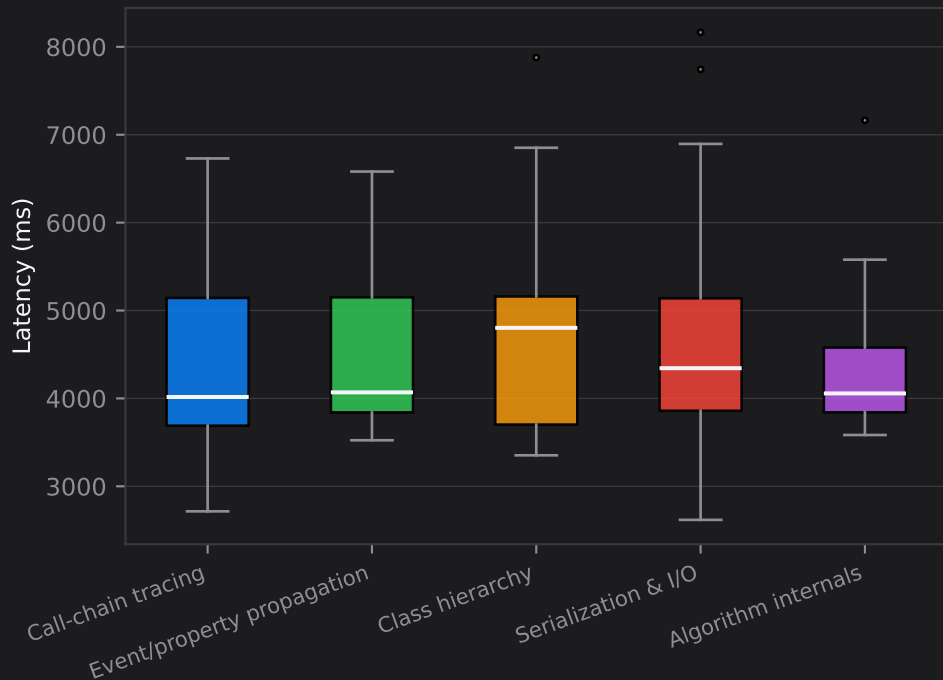
Accuracy@5 and MRR per Seed Dataset



Query Latency Distribution (n=250)

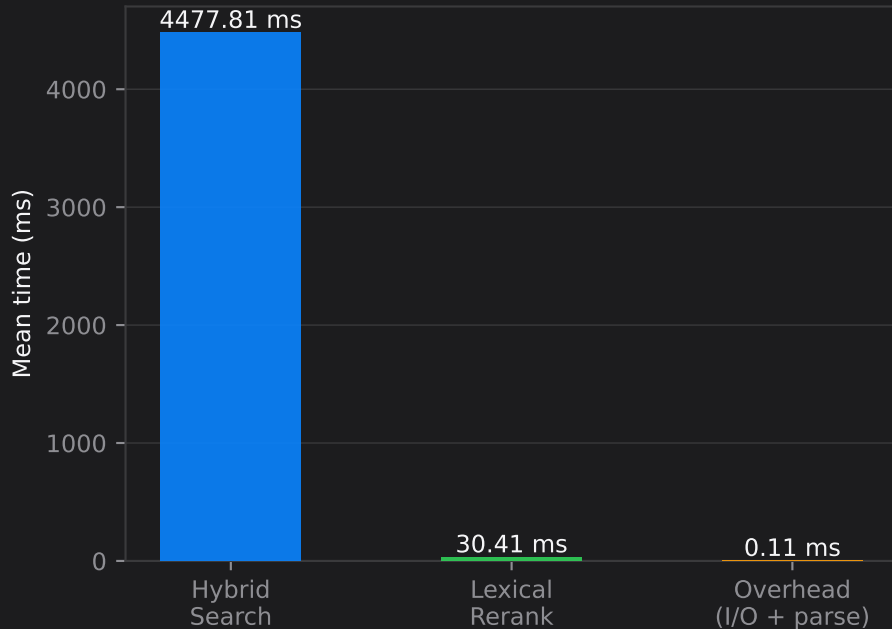


Latency by Seed Dataset

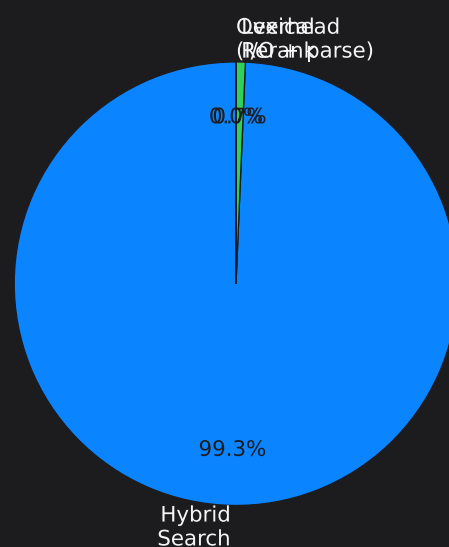


Total mean latency: 4508.34 ms / query

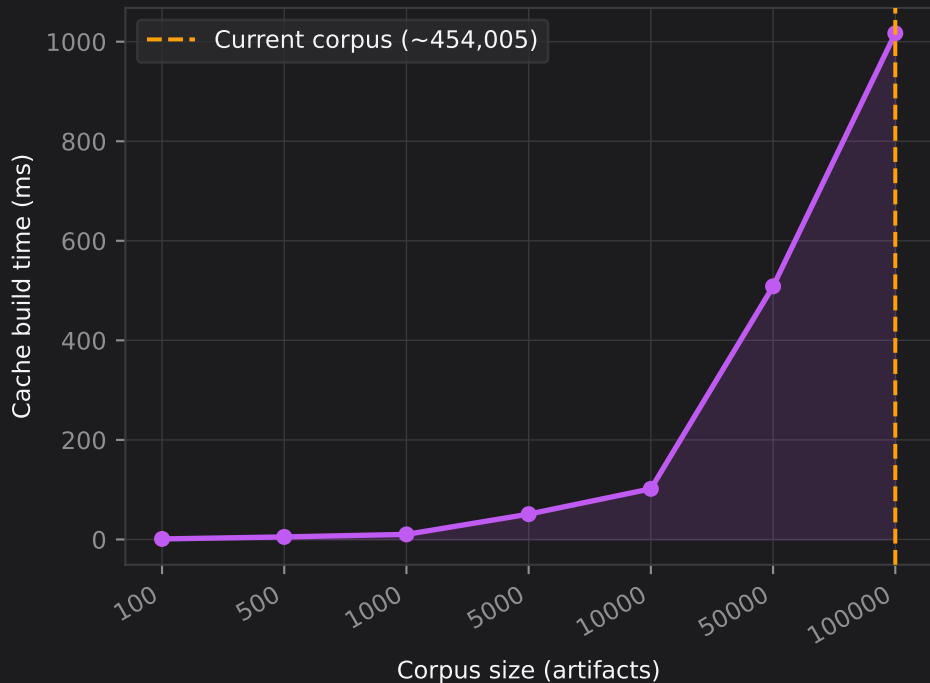
Mean Retrieval Stage Breakdown



Latency Composition



Amortized Numpy Cache Build Time vs Corpus Size



Cache Build: 4617.0 ms | Amortized: 10.17 μ s/artifact

